

WaveMLP: Cross-Receiver-Day RF Fingerprinting with Learnable Complex Wavelets

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Outline

- ① Background & Problem
- ② Motivation
- ③ Method: WaveMLP
- ④ Experiments
- ⑤ Conclusion

Background: RF Fingerprinting (RFF)

What is RFF?

Identify wireless devices through subtle hardware imperfections — a non-cryptographic physical-layer authentication.

The Challenge: Cross-Receiver-Day (CRD)

- Receiver hardware differs between training & deployment
- Capture day (channel, environment) shifts
- Recognition accuracy drops sharply

Existing remedies fall short:

- Channel equalization
- Signal denoising
- Domain adaptation

⇒ Each targets one type of distortion at *fixed* resolution.

Problem Formulation

Signal model. The IQ sequence captured by receiver m on day d :

$$x(k) = \mathcal{G}_{m,d}\{\mathcal{T}_n\{s_n\} * h_{m,d}\}(k) + w_{m,d}(k)$$

- $\mathcal{T}_n\{\cdot\}$: **transmitter fingerprint** (what we want)
- $\xi_{m,d} = \{\mathcal{G}_{m,d}, h_{m,d}, \sigma_{m,d}^2\}$: **scenario factors** (what we want to suppress)

CRD as Domain Generalization. Disjoint receivers and days between train/test:

$$\mathcal{R}_{\text{tr}} \cap \mathcal{R}_{\text{te}} = \emptyset, \quad \mathcal{D}_{\text{tr}} \cap \mathcal{D}_{\text{te}} = \emptyset$$

Goal: learn f_θ on source $\mathcal{P}_S$ that minimizes target risk on $\mathcal{P}_T$.

Motivation: Why Wavelets?

Ben-David bound reveals the source of the CRD gap:

$$\varepsilon_T \leq \varepsilon_S + \frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{P}_S, \mathcal{P}_T) + \lambda^*$$

⇒ We must reduce the divergence between source and target distributions.

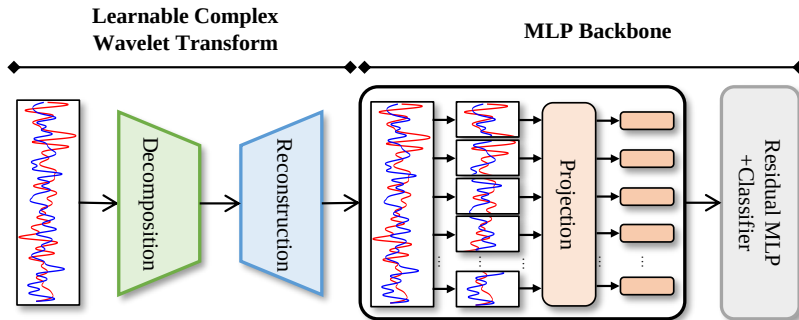
Key observation. Scenario distortions and fingerprints live at *different scales*.

- Channel/receiver effects → often subband-localized
- Device fingerprint → coherent across multiple scales

Our idea

Use a **multi-resolution** time–frequency representation — a learnable wavelet front end — to disentangle the two along the scale axis.

WaveMLP: Overall Architecture



Forward pass:

$$\hat{y} = f_{\theta}(\mathbf{x}) = (g_{\phi} \circ \mathcal{W}_{\psi})(\mathbf{x})$$

- $\mathcal{W}_{\psi}$: **Learnable Complex Wavelet Transform** (LCWT)
- g_{ϕ} : lightweight MLP backbone + classifier
- Trained *end-to-end* with cross-entropy loss

Learnable Complex Wavelet Transform (LCWT)

Design principle. Keep the dyadic filter-bank structure of classical DWT, but *learn* all complex filters from data — no quadrature-mirror constraint.

Complex convolution (preserves IQ algebra):

$$\mathbf{w} \circledast \mathbf{u} = (\mathbf{w}_r \ast \mathbf{u}_r - \mathbf{w}_i \ast \mathbf{u}_i) + j(\mathbf{w}_r \ast \mathbf{u}_i + \mathbf{w}_i \ast \mathbf{u}_r)$$

Analysis (level j):

$$\mathbf{c}^{(j)} = (\mathbf{w}_l^{(j)} \circledast \mathbf{c}^{(j-1)}) \downarrow_2$$

$$\mathbf{d}^{(j)} = (\mathbf{w}_h^{(j)} \circledast \mathbf{c}^{(j-1)}) \downarrow_2$$

Synthesis (level j):

$$\begin{aligned} \mathbf{c}^{(j-1)} &= \tilde{\mathbf{w}}_l^{(j)} \circledast (\mathbf{c}^{(j)} \uparrow_2) \\ &\quad + \tilde{\mathbf{w}}_h^{(j)} \circledast (\mathbf{d}^{(j)} \uparrow_2) \end{aligned}$$

Why decoupled synthesis?

Synthesis filters *complement* (not invert) analysis filters $\Rightarrow$ cross-scale fingerprints retained, subband-localized scenario energy attenuated.

Lightweight MLP Backbone

After LCWT, reconstructed signal $\hat{\mathbf{x}}$ is unfolded:

$$\mathbf{x}_{\text{IQ}} = [\Re(\hat{\mathbf{x}}); \Im(\hat{\mathbf{x}})] \in \mathbb{R}^{2 \times L}$$

- ① **Patch embedding:** 1D conv with patch size P , stride $S \Rightarrow$ token sequence $\mathbf{Z}_0 \in \mathbb{R}^{T \times D}$
- ② **Residual MLP block** with LayerNorm:

$$\mathbf{Z} = \mathbf{Z}_0 + \mathbf{W}_2 \text{GELU}(\mathbf{W}_1 \text{LN}(\mathbf{Z}_0))$$

- ③ **Global Average Pooling** $\rightarrow$ **Fully-connected classifier**

Why MLP, not CNN/Transformer?

LCWT already provides multi-scale features — the back end only needs to *aggregate patch-wise evidence*, so a lightweight MLP suffices.

Experimental Setup

Dataset: WiSig benchmark

- **ManySig:** 6 transmitters $\times$ 12 receivers
- **ManyRx:** 10 transmitters $\times$ 32 receivers
- 100 samples/day over 4 days, sample length 2×256

CRD evaluation protocol:

- Receivers split into 4 disjoint groups $\rightarrow$ 4-fold cross-validation
- Train on first 2 days, test on last 2 days (temporal shift)

Baselines (7 representative methods): TFMix, SigMix, MTL, FS, RIEI, GAN-RXA, AWTD

Training: Adam, $\text{lr} = 10^{-3}$, batch size 128, 200 epochs.

Main Results: Accuracy & Complexity

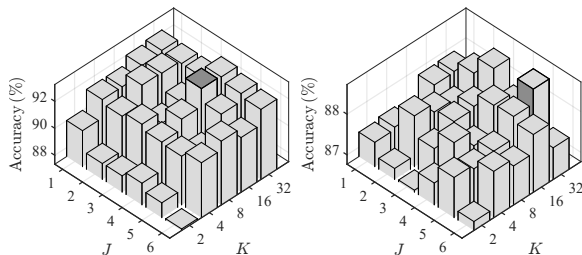
Method	ManySig (Avg.)	ManyRx (Avg.)	Params	FLOPs
TFMix	80.95	73.47	2.59×10^5	2.35×10^7
SigMix	80.01	73.90	2.59×10^5	2.35×10^7
MTL	79.40	72.84	2.86×10^4	2.57×10^6
FS	75.92	68.70	1.37×10^6	1.88×10^9
RIEI	76.07	72.11	4.11×10^6	8.81×10^7
GAN-RXA	78.45	74.33	2.02×10^6	4.08×10^7
AWTD	71.05	86.01	1.12×10^7	3.13×10^7
WaveMLP	93.13	88.68	8.4×10^4	1.3×10^6

- **+12.18%** on ManySig, **+2.67%** on ManyRx vs. best baseline
- **Lowest FLOPs**; 1–2 orders of magnitude lighter than most baselines

Ablation & Visualization

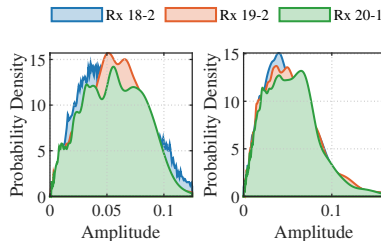
Sensitivity: (J, K)

- Filter length K matters more than depth J
- $J \geq 3$ already covers relevant scales
- Best: $(J, K)=(4, 16)$ on ManySig; $(5, 32)$ on ManyRx



Distribution alignment (PDF)

- 3 receivers (18-2, 19-2, 20-1)
- Avg. pairwise ℓ_1 distance: $0.2371 \rightarrow 0.1072$ (**-54.8%**)
- Visual evidence that LCWT reduces $d_{\mathcal{H}\Delta\mathcal{H}}$



Conclusion & Future Work

Contributions

- Formulated CRD RFF as a **domain generalization** problem
- Proposed **WaveMLP**: a learnable complex wavelet (LCWT) front end + lightweight MLP
- LCWT *decouples* analysis/synthesis filters and lifts the QM constraint, preserving cross-scale fingerprints while suppressing subband scenario energy

Highlights

- **93.13%** on ManySig (+**12.18%**), **88.68%** on ManyRx (+**2.67%**)
- 1–2 orders of magnitude fewer parameters / FLOPs

Future Work

- Extend LCWT to wider modulation families
- Behavior under low-SNR conditions

Thank You!

Q & A

Code: <https://github.com/BeechburgPieStar/WaveMLP-RFF>